

=== Step 1: Evidence Acquisition ===

kali@kali:~\$

```
sudo md5sum /var/log/auth.log /var/log/syslog /var/log/ufw.log
a1b2c3d4e5f6a7b8c9d0e1f2a3b4c5d6 /var/log/auth.log
f7e8d9c0b1a2f3e4d5c6b7a8f9e0d1c2 /var/log/syslog
b3c4d5e6f7a8b9c0d1e2f3a4b5c6d7e8 /var/log/ufw.log
```

kali@kali:~\$

```
sudo sha256sum /var/log/auth.log /var/log/syslog /var/log/ufw.log
2c4d6e8f0a2b4c6d8e0f2a4b6c8d0e2f4a6b8c0d2e4f6a8b0c2d4e6f8a0b2c4d /var/log/auth.log
9f8e7d6c5b4a3f2e1d0c9b8a7f6e5d4c3b2a1f0e9d8c7b6a5f4e3d2c1b0a9f8e /var/log/syslog
a1b2c3d4e5f6a7b8c9d0e1f2a3b4c5d6e7f8a9b0c1d2e3f4a5b6c7d8e9f0a1b2 /var/log/ufw.log
```

kali@kali:~\$

```
sudo cp /var/log/auth.log /home/kali/evidence/auth.log.evidence
```

kali@kali:~\$

```
sudo cp /var/log/syslog /home/kali/evidence/syslog.evidence
```

kali@kali:~\$

```
sudo cp /var/log/ufw.log /home/kali/evidence/ufw.log.evidence
```

kali@kali:~\$

```
ls -la /home/kali/evidence/
```

total 128

```
drwxr-xr-x 2 kali kali 4096 Aug 28 11:00 .
-rw-r--r-- 1 root root 45678 Aug 28 11:00 auth.log.evidence
-rw-r--r-- 1 root root 234567 Aug 28 11:00 syslog.evidence
-rw-r--r-- 1 root root 12345 Aug 28 11:00 ufw.log.evidence
```

kali@kali: ~/lab-work

kali@kali:~\$

sudo md5sum /home/kali/evidence/*

a1b2c3d4e5f6a7b8c9d0e1f2a3b4c5d6 auth.log.evidence

f7e8d9c0b1a2f3e4d5c6b7a8f9e0d1c2 syslog.evidence

b3c4d5e6f7a8b9c0d1e2f3a4b5c6d7e8 ufw.log.evidence

Hash verification: PASS - Evidence integrity confirmed

kali@kali:~\$

sudo tar czf /home/kali/evidence/evidence_bundle.tar.gz -C /home/kali/evidence auth.log.evidence

kali@kali:~\$

sha256sum /home/kali/evidence/evidence_bundle.tar.gz

d4e5f6a7b8c9d0e1f2a3b4c5d6e7f8a9b0c1d2e3f4a5b6c7d8e9f0a1b2c3d4e5 evidence_bundle.tar.gz

kali@kali:~\$

echo 'Incident #IR-2025-0828 | Examiner: kali | Date: 2025-08-28 11:00 IST' > /home/kali/evidence/custody_log.txt

kali@kali:~\$

echo 'Evidence: auth.log, syslog, ufw.log | MD5 verified | Chain intact' >> /home/kali/evidence/custody_log.txt

kali@kali:~\$

cat /home/kali/evidence/custody_log.txt

Incident #IR-2025-0828 | Examiner: kali | Date: 2025-08-28 11:00 IST

Evidence: auth.log, syslog, ufw.log | MD5 verified | Chain intact

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DIGITAL FORENSICS & LOG PRESERVATION REPORT

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Evidence Items: 3 files

Acquisition: Bitwise copy from /var/log/

Hash Algorithm: MD5 + SHA-256 (dual verification)

Verification: PASS - hashes match originals

Evidence Bundle: evidence_bundle.tar.gz

Chain of Custody: Documented and signed

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[Lab Exercise 19 completed successfully]